

BE2100 - Impact of COVID-19 on the U.S. Economy

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1. Introduction

Our project aims to investigate the economic impact of COVID-19 on the U.S. economy. We are looking to understand how key economic indicators such as unemployment rates, inflation rates, retail sales, and GDP have been affected during this period. We aim to provide an in-depth overview of the economic disruptions caused by the pandemic by analyzing trends, variability, and the statistical significance of changes in these indicators. This analysis will offer insights into the overall health of the economy during this challenging period. Through this analysis, we hope to provide a clear and detailed picture of the impact of COVID-19 on the U.S. economy.

2. Approach

We can outline the analysis and solution in the following way to lay out our approach.

1. Data Collection:
 - Collect monthly data for the years 2019 and 2020 on unemployment rates, inflation rates, retail sales, and GDP.
2. Descriptive Statistics:
 - Calculate descriptive statistics for each economic indicator.
3. Visual Analysis:
 - Construct histograms and box plots for each economic indicator to visualize their distribution and identify any outliers or anomalies.
 - Compare the distributions for 2019 and 2020.
4. Statistical Intervals:
 - Calculate confidence intervals for the mean and variance of each economic indicator.
 - Use these intervals to assess the uncertainty and variability in the data.
5. Hypothesis Testing:
 - Perform hypothesis tests to determine if there are statistically significant differences between the 2019 and 2020 data for each economic indicator.
6. Regression Analysis:
 - Conduct simple and multiple linear regression analysis.
7. Conclusion:
 - Summarize and discuss the result of the analysis.
8. Recommendation:

- Provide recommendations based on the analysis.

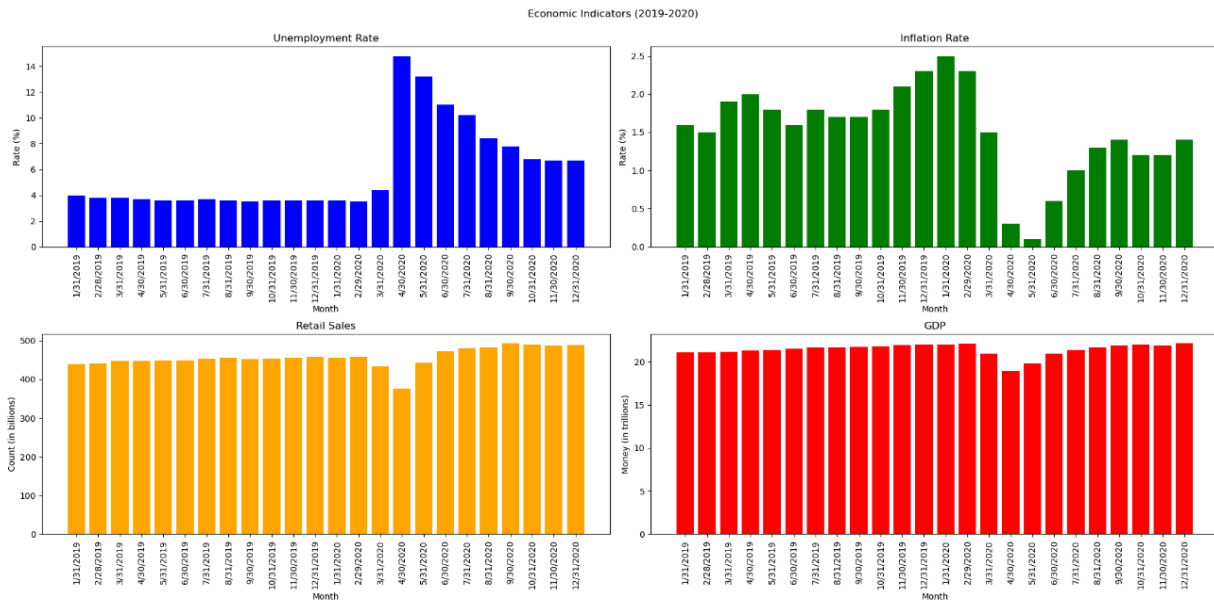
3. Data Collected:

The dataset we collected includes data from sources such as Federal Reserve Economic Data (FRED), U.S. BUREAU OF LABOR STATISTICS, and YCHARTS.

Months (2019 - 2020)	Unemployment Rates	Inflation Rates	Retail Sales (in billions)	GDP (in trillions)
1/31/2019	4	1.6	439.79	21.08
2/28/2019	3.8	1.5	440.88	21.09
3/31/2019	3.8	1.9	446.95	21.14
4/30/2019	3.7	2	447.23	21.28
5/31/2019	3.6	1.8	448.98	21.37
6/30/2019	3.6	1.6	448.99	21.5
7/31/2019	3.7	1.8	453.67	21.64
8/31/2019	3.6	1.7	455.97	21.69
9/30/2019	3.5	1.7	452.05	21.75
10/31/2019	3.6	1.8	453.45	21.77
11/30/2019	3.6	2.1	455.93	21.94
12/31/2019	3.6	2.3	458.92	21.99
1/31/2020	3.6	2.5	456.11	22.02
2/29/2020	3.5	2.3	457.9	22.12
3/31/2020	4.4	1.5	432.93	20.98
4/30/2020	14.8	0.3	375.92	18.94
5/31/2020	13.2	0.1	442.04	19.82
6/30/2020	11	0.6	472.25	20.98
7/31/2020	10.2	1	481.17	21.4
8/31/2020	8.4	1.3	482.39	21.63
9/30/2020	7.8	1.4	492.7	21.91
10/31/2020	6.8	1.2	490.83	22.05
11/30/2020	6.7	1.2	486.83	21.9
12/31/2020	6.7	1.4	488.88	22.13

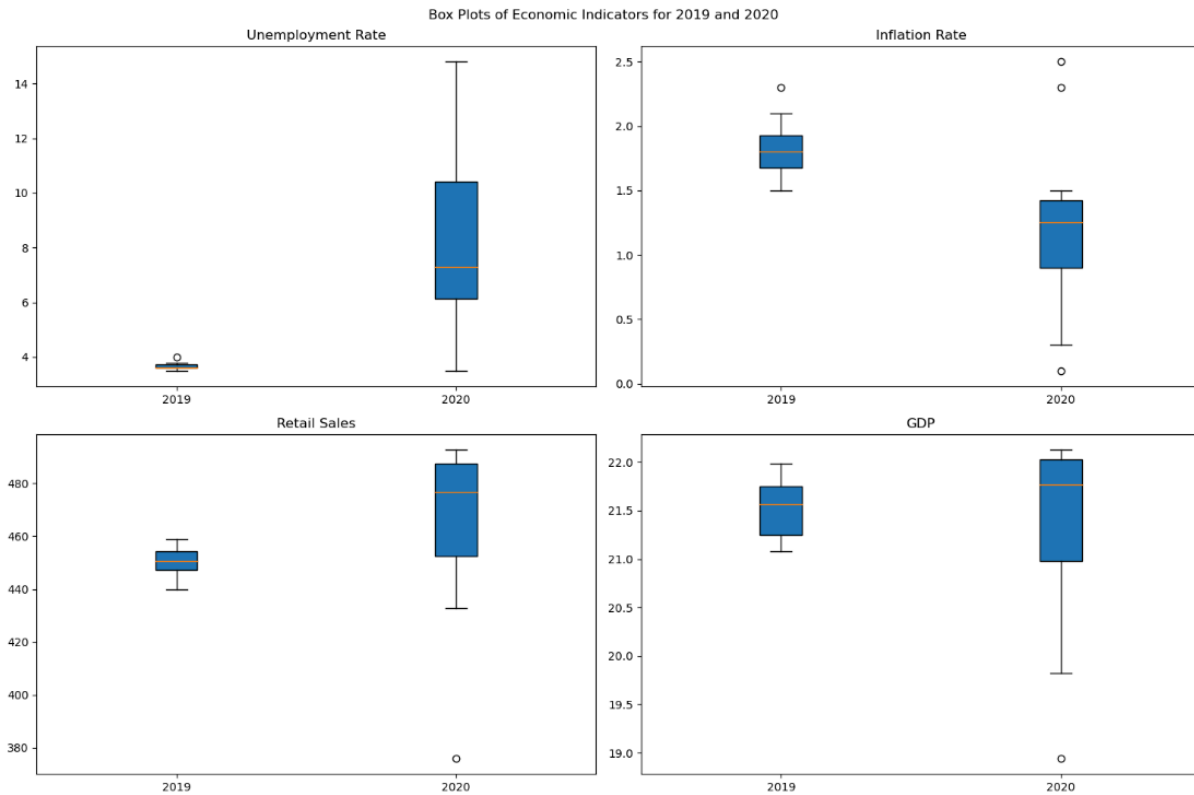
4. Analysis

A) Frequency distributions and / or histogram(s)



- The histogram shows that the unemployment rate remained relatively stable from January 2019 to March 2020, with some fluctuations. There was a sharp increase in April 2020, which correlates with COVID-19.
- The inflation rates appear to fluctuate over the months but show a noticeable dip around the same time as the spike in unemployment. Likely due to reduced economic activity during the pandemic.
- Retail sales seem to take a hit around April 2020 as well, with a notable drop compared to previous months.
- GDP shows a decline starting in the first quarter of 2020, with the lowest point around the second quarter (April to June). The GDP reflects the overall economic output and this drop is indicative of the economic slowdown.

B) Box Plot



Unemployment Rate:

- The median unemployment rate in 2020 is visibly higher than in 2019.
- The range and interquartile range (IQR) are also significantly larger in 2020, indicating increased variability in unemployment rates during the COVID-19 pandemic.
- There are no outliers in 2019, but there is a presence of outliers in 2020, which suggests that in some months, the unemployment rate was exceptionally high.

Inflation Rate:

- The median inflation rate is lower in 2020 compared to 2019.
- The IQR for 2020 is higher than for 2019, indicating greater variability in inflation rates during the COVID-19 pandemic.
- Outliers are present for both years, with 2020 showing particularly high values well above the upper quartile, which suggests instances of unexpected inflation spikes.

Retail Sales:

- There is an increase in the median retail sales from 2019 to 2020, which could be attributed to a shift in consumer spending patterns.

- The range and IQR are more broad in 2020, showing more significant fluctuations in retail sales, likely due to the economic impact.
- The presence of outliers in 2020 may indicate months of either a significant decrease or a surge in retail sales.

GDP:

- The median GDP appears to be slightly higher in 2020 than in 2019, but the range and IQR are both larger in 2020, indicating greater economic variability.
- The outlier in 2020 suggests an exceptionally low GDP value for at least one month, possibly during the peak economic recession due to the COVID-19 pandemic.

Outliers, particularly in unemployment and GDP, may not be anomalies but rather actual representations of the extreme economic conditions experienced during specific months.

C) Summary of descriptive statistics

	Year	2019	2020
Unemployment Rates	mean	3.675000	8.091667
	std	0.135680	3.625781
	var	0.018409	13.146288
	min	3.500000	3.500000
	median	3.600000	7.300000
	max	4.000000	14.800000
Inflation Rates	mean	1.816667	1.233333
	std	0.228963	0.706249
	var	0.052424	0.498788
	min	1.500000	0.100000
	median	1.800000	1.250000
	max	2.300000	2.500000
Retail Sales (in billions)	mean	450.234167	463.329167
	std	5.921782	33.957943
	var	35.067499	1153.141863
	min	439.790000	375.920000
	median	450.520000	476.710000
	max	458.920000	492.700000
GDP (in trillions)	mean	21.520000	21.323333
	std	0.324654	1.011366
	var	0.105400	1.022861
	min	21.080000	18.940000
	median	21.570000	21.765000
	max	21.990000	22.130000

The descriptive statistics indicate that the unemployment rate ranged from 3.5% to 14.8% during the period, with an average rate of approximately 5.9%. The relatively low standard deviation of around 0.13 for 2019 and 3.63 for 2020 suggests that the unemployment rate was fairly stable in 2019, but had significant change in 2020, likely due to the impact of COVID-19. The average inflation rate decreased from 1.817% in 2019 to 1.233% in 2020, which could reflect a slowing economy. The variability in inflation rates, as measured by the standard

deviation, increased from 0.229% in 2019 to 0.706% in 2020, indicating more fluctuation in prices during the pandemic. Retail sales showed a general increasing trend from around 439.79 billion to 492.7 billion, with some fluctuations. The standard deviation of approximately 5.92 billion for 2019 and 33.96 billion for 2020 suggests that there was significant variability in retail sales, possibly influenced by economic conditions. Similar to Retail Sales, the GDP exhibited a slight increasing trend from 21.08 trillion to 22.13 trillion, with some fluctuations. The standard deviation of around 0.32 trillion for 2019 and 1.01 trillion for 2020 indicates that there was considerable variability in GDP, possibly reflecting changes in economic activity.

B) Statistical Interval

Using the sample data for 2019 and 2020, confidence intervals for the mean of each economic indicator (unemployment rate, inflation rate, retail sales, and GDP) is calculated to estimate the range in which the population mean is likely to lie.

$$P\left(\bar{x} - t_{\frac{\alpha}{2}, n-1} \left(\frac{s}{\sqrt{n}}\right) \leq \mu \leq \bar{x} + t_{\frac{\alpha}{2}, n-1} \left(\frac{s}{\sqrt{n}}\right)\right) = 0.95$$

$$t_{\frac{\alpha}{2}, n-1} = t_{\frac{0.05}{2}, 12-1} = 2.201$$

1.

a) Unemployment Rates 2019:

$$P\left(3.675 - 2.201 \frac{0.135680}{\sqrt{12}} \leq \mu \leq 3.675 + 2.201 \frac{0.135680}{\sqrt{12}}\right) = 0.95$$

$$P(3.588793 \leq \mu \leq 3.761207) = 0.95$$

b) Unemployment Rates 2020:

$$P\left(8.092 - 2.201 \frac{3.625781}{\sqrt{12}} \leq \mu \leq 8.092 + 2.201 \frac{3.625781}{\sqrt{12}}\right) = 0.95$$

$$P(5.787955 \leq \mu \leq 10.395378) = 0.95$$

2.

a) Inflation Rates 2019:

$$P\left(1.817 - 2.201 \frac{0.228963}{\sqrt{12}} \leq \mu \leq 1.817 + 2.201 \frac{0.228963}{\sqrt{12}}\right) = 0.95$$

$$P(1.67119 \leq \mu \leq 1.962143) = 0.95$$

b) Inflation Rates 2020:

$$P\left(1.233 - 2.201 \frac{0.706249}{\sqrt{12}} \leq \mu \leq 1.233 + 2.201 \frac{0.706249}{\sqrt{12}}\right) = 0.95$$

$$P(0.784604 \leq \mu \leq 1.682063) = 0.95$$

3.

a) Retail Sales 2019:

$$P\left(450.234 - 2.201 \frac{5.921782}{\sqrt{12}} \leq \mu \leq 450.234 + 2.201 \frac{5.921782}{\sqrt{12}}\right) = 0.95$$

$$P(446.471646 \leq \mu \leq 453.996687) = 0.95$$

b) Retail Sales 2020:

$$P\left(463.329 - 2.201 \frac{33.957943}{\sqrt{12}} \leq \mu \leq 463.329 + 2.201 \frac{33.957943}{\sqrt{12}}\right) = 0.95$$

$$P(441.753319 \leq \mu \leq 484.905014) = 0.95$$

4.

a) GDP 2019:

$$P\left(21.52 - 2.201 \frac{0.325}{\sqrt{12}} \leq \mu \leq 21.52 + 2.201 \frac{0.325}{\sqrt{12}}\right) = 0.95$$

$$P(21.313725 \leq \mu \leq 21.726275) = 0.95$$

b) GDP 2020:

$$P\left(21.323 - 2.201 \frac{1.011366}{\sqrt{12}} \leq \mu \leq 21.323 + 2.201 \frac{1.011366}{\sqrt{12}}\right) = 0.95$$

$$P(20.680742 \leq \mu \leq 21.965924) = 0.95$$

- The wider confidence interval in 2020 [5.788%, 10.395%] compared to 2019 [3.589%, 3.761%] reflects the significant increase in uncertainty and variability in unemployment rates.
- The shift in the confidence interval from 2019 [1.671%, 1.962%] to 2020 [0.785%, 1.682%] indicates a general decrease in inflation rates during the pandemic, with a lower upper bound in 2020.
- The increase in the upper bound of the confidence interval from 2019 [446.472 billion, 453.997 billion] to 2020 [441.755 billion, 484.905 billion] suggests potential spikes in retail sales despite the overall economic impact of the pandemic.
- The wider confidence interval in 2020 [20.681 trillion, 21.966 trillion] compared to 2019 [21.314 trillion, 21.726 trillion] highlights the increased uncertainty and the economic contraction experienced during the pandemic.

Use the chi-squared distribution to calculate the confidence intervals for the variance of each economic indicator for 2019 and 2020. The formula for a 95% confidence interval for variance being used (σ^2) is:

$$P\left(\frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2}, n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}}\right) = 0.95$$

$$\alpha = 1 - 0.95 = 0.05$$

$$\chi^2_{\frac{\alpha}{2}, n-1} = \chi^2_{\frac{0.05}{2}, 12-1} = 21.92$$

$$\chi^2_{1-\frac{\alpha}{2}, n-1} = \chi^2_{1-\frac{0.05}{2}, 12-1} = 3.82$$

1.

a) Unemployment Rate 2019:

$$P\left(\frac{(12-1)13.146288}{21.92} \leq \sigma^2 \leq \frac{(12-1)13.146288}{3.82}\right) = 0.95$$

$$P(0.009238 \leq \sigma^2 \leq 0.05307) = 0.95$$

b) Unemployment Rate 2020:

$$P\left(\frac{(12-1)13.146288}{21.92} \leq \sigma^2 \leq \frac{(12-1)13.146288}{3.82}\right) = 0.95$$

$$P(6.597119 \leq \sigma^2 \leq 37.897984) = 0.95$$

2.

a) Inflation Rate 2019:

$$P\left(\frac{(12-1)0.052424}{21.92} \leq \sigma^2 \leq \frac{(12-1)0.052424}{3.82}\right) = 0.95$$

$$P(0.026308 \leq \sigma^2 \leq 0.151128) = 0.95$$

b) Inflation Rate 2020:

$$P\left(\frac{(12-1)0.498788}{21.92} \leq \sigma^2 \leq \frac{(12-1)0.498788}{3.82}\right) = 0.95$$

$$P(0.250304 \leq \sigma^2 \leq 1.437901) = 0.95$$

3.

a) Retail Sales 2019:

$$P\left(\frac{(12-1)35.067499}{21.92} \leq \sigma^2 \leq \frac{(12-1)35.067499}{3.82}\right) = 0.95$$

$$P(17.597702 \leq \sigma^2 \leq 101.092228) = 0.95$$

b) Retail Sales 2020:

$$P\left(\frac{(12-1)1153.141863}{21.92} \leq \sigma^2 \leq \frac{(12-1)1153.141863}{3.82}\right) = 0.95$$

$$P(578.673904 \leq \sigma^2 \leq 3324.265558) = 0.95$$

4.

c) GDP 2019:

$$P\left(\frac{(12-1)0.105400}{21.92} \leq \sigma^2 \leq \frac{(12-1)0.105400}{3.82}\right) = 0.95$$

$$P(0.052892 \leq \sigma^2 \leq 0.303846) = 0.95$$

d) GDP 2020:

$$P\left(\frac{(12-1)1.022861}{21.92} \leq \sigma^2 \leq \frac{(12-1)1.022861}{3.82}\right) = 0.95$$

$$P(0.513296 \leq \sigma^2 \leq 2.948692) = 0.95$$

- The significantly wider confidence interval for variance in 2020 [6.597, 37.898] compared to 2019 [0.009, 0.053] reflects the dramatic increase in variability and uncertainty in unemployment rates due to the COVID-19 pandemic.
- The increase in the confidence interval for variance from 2019 [0.026, 0.151] to 2020 [0.250, 1.438] indicates heightened variability in inflation rates.
- The substantial widening of the confidence interval for variance from 2019 [17.598, 101.092] to 2020 [578.694, 3324.266] captures the increased variability and unpredictability in retail sales during the pandemic.
- The broader confidence interval for variance in 2020 [0.513, 2.949] compared to 2019 [0.053, 0.304] highlights the increased variability in GDP, reflecting the economic impact.

C) Tests for Significance

Significance tests will help confirm whether the differences observed between 2019 and 2020 are statistically significant or could have occurred by chance. By testing for significant differences in means and variances, we can quantitatively assess the extent of the pandemic's impact on various aspects of the economy.

- Mean:
- For the unemployment rate, where there is no overlap in the statistical intervals, there is no need to test significance since the difference is significant.

- Inflation Rates, Retail Sales, and GDP have overlap and the significance needs to be tested.

1. Inflation Rate:

Null Hypothesis: $H_0 : \mu_{2019} = \mu_{2020}$

Alternative Hypothesis: $H_a : \mu_{2019} \neq \mu_{2020}$

Test Statistic:
$$t_0 = \frac{\bar{x}_{2019} - \bar{x}_{2020}}{\sqrt{\frac{s_{2019}^2}{n_{2019}} + \frac{s_{2020}^2}{n_{2020}}}}$$

$$t_0 = \frac{1.816667 - 1.233333}{\sqrt{\frac{0.052424}{12} + \frac{0.498788}{12}}} = 2.722$$

Reject H_0 if: $|t_0| > t_{\frac{\alpha}{2}, n-1}$

$$\alpha = 1 - 0.95 = 0.05$$

$$t_{\frac{\alpha}{2}, n-1} = t_{0.025, 11} = 2.201$$

- Since the absolute value of the test statistic ($|t_0| = 2.722$) is greater than the critical t-value (2.201), the null hypothesis H_0 is rejected. There is a statistically significant difference in the mean Inflation Rate between 2019 and 2020 at the 5% significance level.

2. Retail Sales:

Null Hypothesis: $H_0 : \mu_{2019} = \mu_{2020}$

Alternative Hypothesis: $H_a : \mu_{2019} \neq \mu_{2020}$

Test Statistic:
$$t_0 = \frac{\bar{x}_{2019} - \bar{x}_{2020}}{\sqrt{\frac{s_{2019}^2}{n_{2019}} + \frac{s_{2020}^2}{n_{2020}}}}$$

$$t_0 = \frac{450.234167 - 463.329167}{\sqrt{\frac{35.067499}{12} + \frac{1153.141863}{12}}} = -1.316$$

Reject H_0 if: $|t_0| > t_{\frac{\alpha}{2}, n-1}$

$$\alpha = 1 - 0.95 = 0.05$$

$$t_{\frac{\alpha}{2}, n-1} = t_{0.025, 11} = 2.201$$

- Since the absolute value of the test statistic ($|t_0| = 1.316$) is less than the critical t-value (2.201), the null hypothesis H_0 is failed to be rejected. There is no statistically significant difference in the mean Retail Sales between 2019 and 2020 at the 5% significance level. This suggests that despite the COVID-19 pandemic, the overall level of Retail Sales may not have changed significantly.

3. GDP:

Null Hypothesis: $H_0 : \mu_{2019} = \mu_{2020}$

Alternative Hypothesis: $H_a : \mu_{2019} \neq \mu_{2020}$

Test Statistic:
$$t_0 = \frac{\bar{x}_{2019} - \bar{x}_{2020}}{\sqrt{\frac{s_{2019}^2}{n_{2019}} + \frac{s_{2020}^2}{n_{2020}}}}$$

$$t_0 = \frac{21.520 - 21.323333}{\sqrt{\frac{0.105400}{12} + \frac{1.022861}{12}}} = 0.641$$

Reject H_0 if: $|t_0| > t_{\frac{\alpha}{2}, n-1}$

$$\alpha = 1 - 0.95 = 0.05$$

$$t_{\frac{\alpha}{2}, n-1} = t_{0.025, 11} = 2.201$$

- Since the absolute value of the test statistic ($|t_0| = 0.641$) is less than the critical t-value (2.201), the null hypothesis H_0 is failed to be rejected. This suggests that the overall level of GDP may not have changed significantly due to COVID-19.

- Variance:

1. Unemployment Rate:

Null Hypothesis: $H_0 : \mu_{2019} = \mu_{2020}$

Alternative Hypothesis: $H_a : \mu_{2019} \neq \mu_{2020}$

Test Statistic: $f_0 = \frac{\text{larger variance}}{\text{smaller variance}} = \frac{s_{2020}^2}{s_{2019}^2}$

$$f_0 = \frac{13.146288}{0.018409} = 714.123$$

Reject H_0 if $f_0 > f_{\frac{\alpha}{2}, n-1, n-1}$

$$\alpha = 1 - 0.95 = 0.05$$

$$f_{\frac{\alpha}{2}, n-1, n-1} = f_{0.025, 11, 11} = 3.4737$$

- Since the test statistic ($f = 714.123$) is greater than the critical value (3.4737) the null hypothesis H_0 is rejected.

2. Inflation Rate:

Null Hypothesis: $H_0 : \mu_{2019} = \mu_{2020}$

Alternative Hypothesis: $H_a : \mu_{2019} \neq \mu_{2020}$

Test Statistic:
$$f_0 = \frac{\text{larger variance}}{\text{smaller variance}} = \frac{s_{2020}^2}{s_{2019}^2}$$

$$f_0 = \frac{0.498788}{0.052424} = 9.515$$

Reject H_0 if
$$f_0 > f_{\frac{\alpha}{2}, n-1, n-1}$$

$$\alpha = 1 - 0.95 = 0.05$$

$$f_{\frac{\alpha}{2}, n-1, n-1} = f_{0.025, 11, 11} = 3.4737$$

- Since the test statistic ($f = 9.515$) is greater than the critical value (3.4737) the null hypothesis H_0 is rejected.

3. Retail Sales:

Null Hypothesis:
$$H_0 : \mu_{2019} = \mu_{2020}$$

Alternative Hypothesis:
$$H_a : \mu_{2019} \neq \mu_{2020}$$

Test Statistic:
$$f_0 = \frac{\text{larger variance}}{\text{smaller variance}} = \frac{s_{2020}^2}{s_{2019}^2}$$

$$f_0 = \frac{1153.141863}{35.067499} = 32.883$$

Reject H_0 if
$$f_0 > f_{\frac{\alpha}{2}, n-1, n-1}$$

$$\alpha = 1 - 0.95 = 0.05$$

$$f_{\frac{\alpha}{2}, n-1, n-1} = f_{0.025, 11, 11} = 3.4737$$

- Since the test statistic ($f = 32.883$) is greater than the critical value (3.4737) the null hypothesis H_0 is rejected.

4. GDP:

Null Hypothesis: $H_0 : \mu_{2019} = \mu_{2020}$

Alternative Hypothesis: $H_a : \mu_{2019} \neq \mu_{2020}$

Test Statistic: $f_0 = \frac{\text{larger variance}}{\text{smaller variance}} = \frac{s_{2020}^2}{s_{2019}^2}$

$$f_0 = \frac{1.022861}{0.105400} = 9.705$$

Reject H_0 if $f_0 > f_{\frac{\alpha}{2}, n-1, n-1}$

$$\alpha = 1 - 0.95 = 0.05$$

$$f_{\frac{\alpha}{2}, n-1, n-1} = f_{0.025, 11, 11} = 3.4737$$

- Since the test statistic ($f = 9.705$) is greater than the critical value (3.4737) the null hypothesis H_0 is rejected.

D) Regression Analysis

- GDP vs Retail Sales - Simple Linear Regression
 - Dependent Variable: GDP (in trillions)
 - Independent Variable: Retail Sales (in billions)

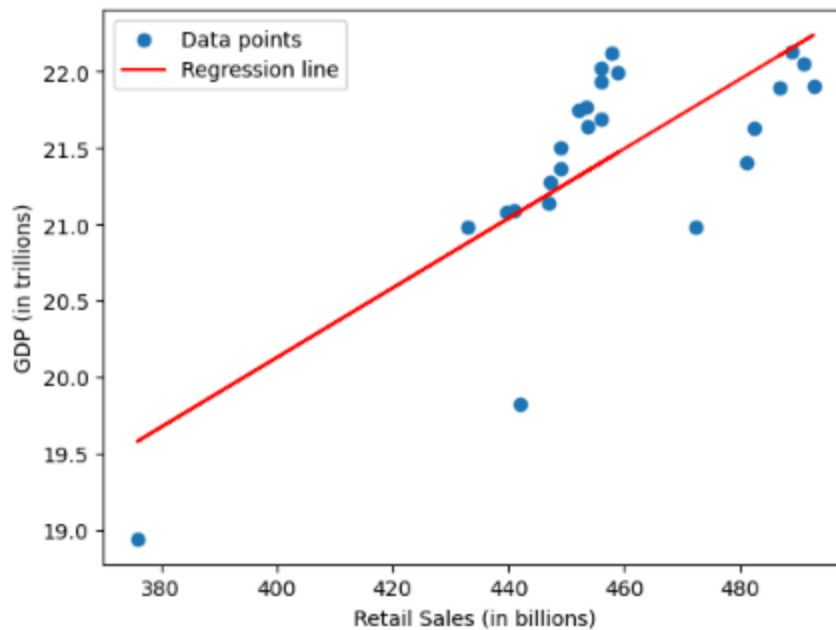

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OLS Regression Results
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Dep. Variable:    GDP (in trillions)    R-squared:        0.579
Model:            OLS                    Adj. R-squared:    0.560
Method:            Least Squares         F-statistic:       30.31
Date:              Thu, 04 Apr 2024      Prob (F-statistic): 1.57e-05
Time:              02:10:04              Log-Likelihood:    -15.970
No. Observations: 24                    AIC:               35.94
Df Residuals:      22                    BIC:               38.30
Df Model:           1
Covariance Type:   nonrobust
=====
                    coef    std err          t      P>|t|      [0.025    0.975]
-----
const              11.0099      1.894      5.813     0.000      7.082    14.938
Retail Sales (in billions)  0.0228      0.004      5.505     0.000      0.014     0.031
=====
Omnibus:           4.107    Durbin-Watson:      0.328
Prob(Omnibus):     0.128    Jarque-Bera (JB):    2.693
Skew:              -0.808    Prob(JB):            0.260
Kurtosis:          3.281    Cond. No.            8.63e+03
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Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 8.63e+03. This might indicate that there are strong multicollinearity or other numerical problems.



- About 57.9% of the variation in GDP is explained by Retail Sales.
- Adjusts the R-squared for the number of predictors in the model, still indicating a good fit.
- The model is statistically significant (p-value from F-statistic is very small).

- The red line in the scatter plot represents the best-fit line from the regression analysis, showing a positive slope. Which means that as Retail Sales increase, GDP also tends to increase.

Further investigation is needed to confirm these findings, possibly including more variables in the analysis.

- GDP vs Retail Sales, Unemployment Rates, Inflation Rates - Multiple Linear Regression
- Dependent Variable: GDP (in trillions)
- Independent Variable: Retail Sales (in billions), Inflation Rates, Unemployment Rates

OLS Regression Results

Dep. Variable:	GDP (in trillions)	R-squared:	0.955
Model:	OLS	Adj. R-squared:	0.948
Method:	Least Squares	F-statistic:	140.5
Date:	Fri, 05 Apr 2024	Prob (F-statistic):	1.32e-13
Time:	03:05:18	Log-Likelihood:	10.773
No. Observations:	24	AIC:	-13.55
Df Residuals:	20	BIC:	-8.833
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	11.5400	0.732	15.758	0.000	10.012	13.068
Retail Sales (in billions)	0.0204	0.001	14.153	0.000	0.017	0.023
Unemployment Rates	-0.0423	0.025	-1.723	0.100	-0.094	0.009
Inflation Rates	0.5473	0.140	3.911	0.001	0.255	0.839

Omnibus:	2.098	Durbin-Watson:	1.057
Prob(Omnibus):	0.350	Jarque-Bera (JB):	1.548
Skew:	-0.614	Prob(JB):	0.461
Kurtosis:	2.801	Cond. No.	9.73e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 9.73e+03. This might indicate that there are strong multicollinearity or other numerical problems.

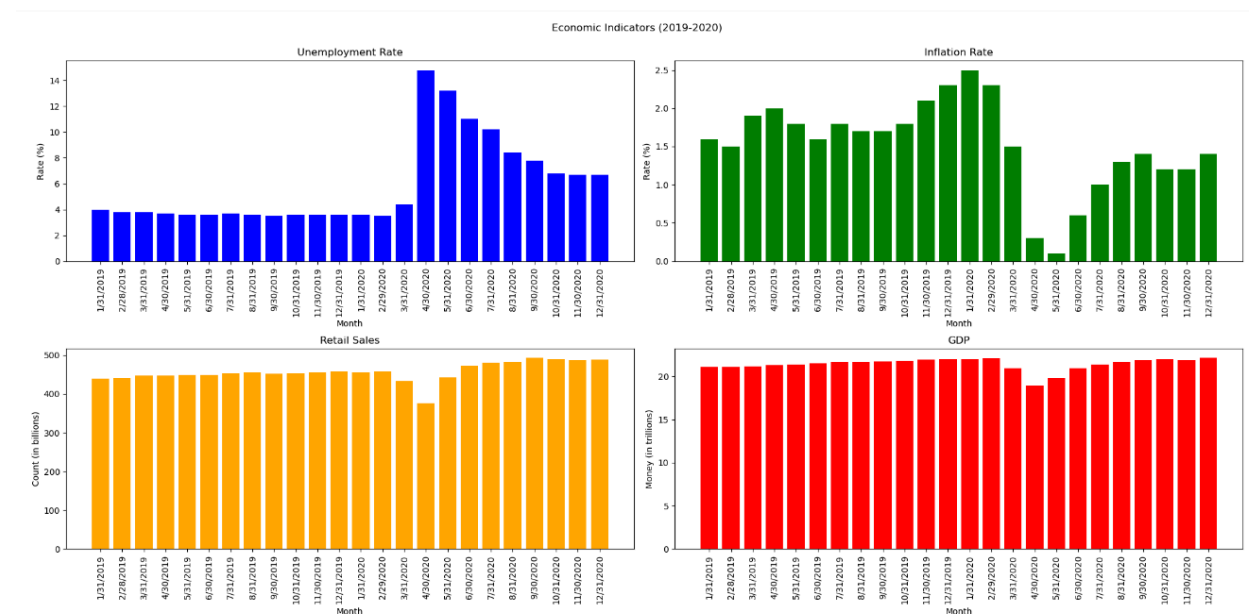
- The model has a very high R-squared value of 0.955, indicating that 95.5% of the variability in GDP is explained by the model's inputs: Retail Sales, Unemployment Rates, and Inflation Rates.

- The adjusted R-squared value is 0.948, which takes into account the number of predictors in the model and suggests a good fit.
- The model is statistically significant (p-value from F-statistic is very small).
- Retail Sales and Inflation Rates both have positive coefficients ($p < 0.05$), which is statistically significant.

The multiple linear regression analysis suggests a significant relationship between GDP and both Retail Sales and Inflation Rates, but not with the Unemployment Rate. This suggests that the model does not provide strong evidence of the Unemployment Rate influencing GDP within this dataset.

5. Conclusion

In our analysis, we focused on the economic impact of the COVID-19 pandemic on the U.S. economy, specifically examining key economic indicators such as Unemployment Rates, Inflation Rates, Retail Sales, and GDP.



Our findings indicate a significant shift in these economic indicators around April 2020, which corresponds with the emergence of the COVID-19 pandemic. The unemployment rates, which had been relatively stable, saw a sharp increase. Similarly, retail sales experienced a notable

drop, and GDP began to decline in the first quarter of 2020, reaching its lowest point in the second quarter.

We conducted significance tests to determine whether these changes were statistically significant or could have occurred by chance. For the unemployment rate, the tests revealed a statistically significant difference between the means of 2019 and 2020, indicating that the increase in unemployment rates during the pandemic was not due to random variation. The tests for inflation rates also showed a significant difference in the means between the two years, suggesting that the pandemic influenced inflation dynamics.

Furthermore, our simple linear regression analysis indicates that about 57.9% of the variation in GDP is explained by Retail Sales. The multiple linear regression analysis model has a very high R-squared value of 0.955, suggesting that 95.5% of the variability in GDP is explained by the model's inputs: Retail Sales, Unemployment Rates, and Inflation Rates.

In conclusion, our analysis highlights the profound economic impact of the COVID-19 pandemic on the U.S. economy. The increased variability and significant changes in unemployment rates, inflation rates, retail sales, and GDP highlight the uncertainty and hardships encountered during this time period. Understanding these patterns is critical for policymakers, economists, and businesses as they manage the current recovery and plan for future economic disruptions.

6. Recommendation:

We recommend further research into the long-term effects and a comparative analysis with other countries to understand the global impact of Covid-19. To improve future analyses, we suggest collecting more detailed data, including data on different sectors, demographic groups, and regions or states. We also recommend looking at the impact of digital transformation on economic sustainability during COVID-19. This could include researching the development of e-commerce, remote work, and online education, among others.

7. Sources:

Data was collected from these following sources:

<https://fred.stlouisfed.org/series/CPIAUCSL>

<https://www.bls.gov/charts/employment-situation/civilian-unemployment-rate.htm>

https://ycharts.com/indicators/us_monthly_gdp

https://ycharts.com/indicators/us_retail_sales

Appendix

▼ Histogram

```
In [2]: import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.ticker import FuncFormatter

data = pd.read_csv('EconomyData.csv')

months = list(range(1, len(data) + 1))
unemployment_rates = data['Unemployment Rates'].tolist()
monthly_inflation_rates = data['Inflation Rates'].tolist()
unemployed_people = data['Retail Sales (in billions)'].tolist()
gdp = data['GDP (in trillions)'].tolist()

fig, axs = plt.subplots(2, 2, figsize=(20, 10))
fig.suptitle('Economic Indicators (2019-2020)')

axs[0, 0].bar(months, unemployment_rates, color='blue')
axs[0, 0].set_title('Unemployment Rate')
axs[0, 0].set_xlabel('Month')
axs[0, 0].set_ylabel('Rate (%)')
axs[0, 0].set_xticks(months)
axs[0, 0].set_xticklabels(data['Months (2019 - 2020)'], rotation=90)

axs[0, 1].bar(months, monthly_inflation_rates, color='green')
axs[0, 1].set_title('Inflation Rate')
axs[0, 1].set_xlabel('Month')
axs[0, 1].set_ylabel('Rate (%)')
axs[0, 1].set_xticks(months)
axs[0, 1].set_xticklabels(data['Months (2019 - 2020)'], rotation=90)

axs[1, 0].bar(months, unemployed_people, color='orange')
axs[1, 0].set_title('Retail Sales')
axs[1, 0].set_xlabel('Month')
axs[1, 0].set_ylabel('Count (in billions)')
axs[1, 0].set_xticks(months)
axs[1, 0].set_xticklabels(data['Months (2019 - 2020)'], rotation=90)
```

Boxplot

```
In [2]: data['Year'] = pd.to_datetime(data['Months (2019 - 2020)']).dt.year
data_2019 = data[data['Year'] == 2019]
data_2020 = data[data['Year'] == 2020]

fig, axs = plt.subplots(2, 2, figsize=(15, 10))
fig.suptitle('Box Plots of Economic Indicators for 2019 and 2020')

axs[0, 0].boxplot([data_2019['Unemployment Rates'], data_2020['Unemployment Rates']], patch_artist=True)
axs[0, 0].set_title('Unemployment Rate')
axs[0, 0].set_xticklabels(['2019', '2020'])

axs[0, 1].boxplot([data_2019['Inflation Rates'], data_2020['Inflation Rates']], patch_artist=True)
axs[0, 1].set_title('Inflation Rate')
axs[0, 1].set_xticklabels(['2019', '2020'])

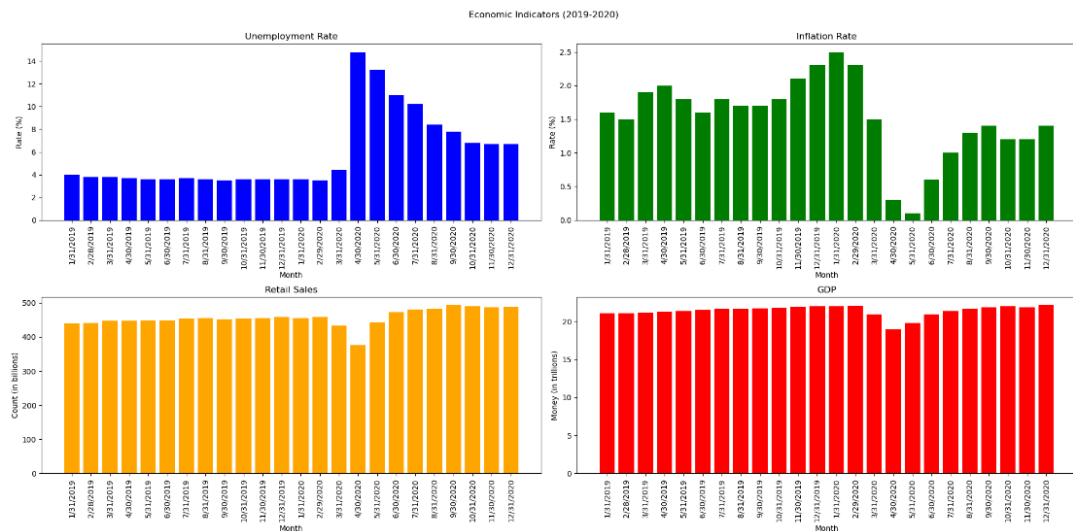
axs[1, 0].boxplot([data_2019['Retail Sales (in billions)'], data_2020['Retail Sales (in billions)']], patch_artist=True)
axs[1, 0].set_title('Retail Sales')
axs[1, 0].set_xticklabels(['2019', '2020'])

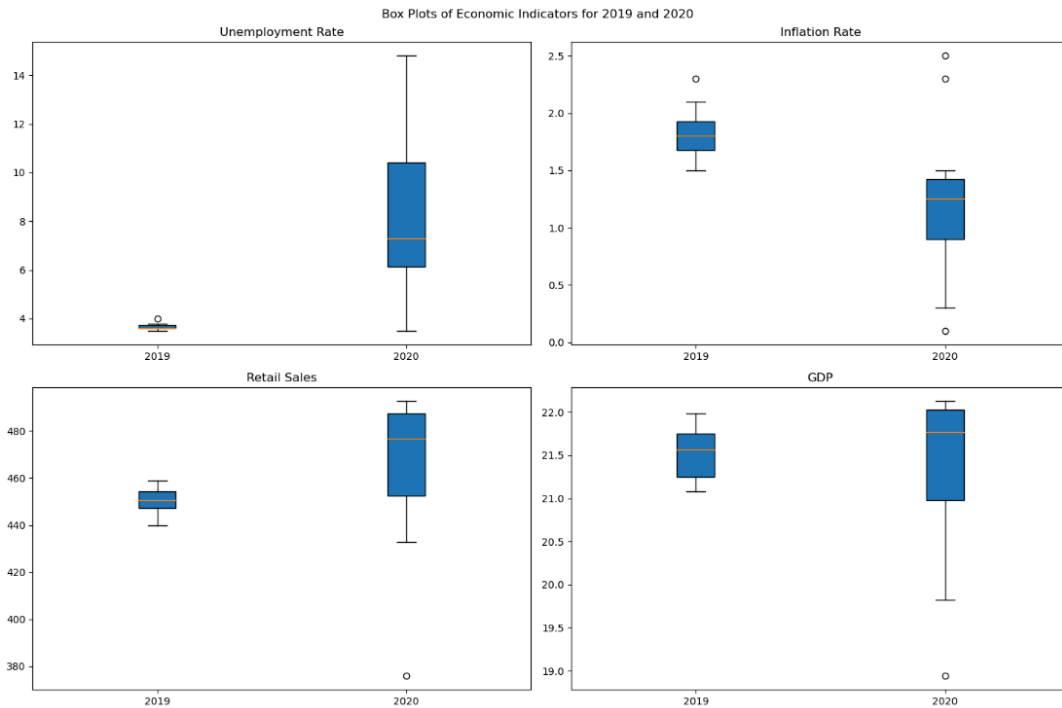
axs[1, 1].boxplot([data_2019['GDP (in trillions)'], data_2020['GDP (in trillions)']], patch_artist=True)
axs[1, 1].set_title('GDP')
axs[1, 1].set_xticklabels(['2019', '2020'])

plt.tight_layout()
plt.show()
```

```
axs[1, 1].bar(months, gdp, color='red')
axs[1, 1].set_title('GDP')
axs[1, 1].set_xlabel('Month')
axs[1, 1].set_ylabel('Money (in trillions)')
axs[1, 1].set_xticks(months)
axs[1, 1].set_xticklabels(data['Months (2019 - 2020)'], rotation=90)

plt.tight_layout()
plt.subplots_adjust(top=0.92)
plt.show()
```





Descriptive Statistics

```
In [3]: import pandas as pd

data = pd.read_csv('EconomyData.csv')

data['Year'] = pd.to_datetime(data['Months (2019 - 2020)']).dt.year

descriptive_stats = data.groupby('Year').agg({
    'Unemployment Rates': ['mean', 'std', 'var', 'min', 'median', 'max'],
    'Inflation Rates': ['mean', 'std', 'var', 'min', 'median', 'max'],
    'Retail Sales (in billions)': ['mean', 'std', 'var', 'min', 'median', 'max'],
    'GDP (in trillions)': ['mean', 'std', 'var', 'min', 'median', 'max']
}).transpose()

for column in ['Unemployment Rates', 'Inflation Rates', 'Retail Sales (in billions)', 'GDP (in trillions)']:
    descriptive_stats.loc[(column, 'iqr'), :] = (
        data.groupby('Year')[column].quantile(0.75) - data.groupby('Year')[column].quantile(0.25)
    )

for column in ['Unemployment Rates', 'Inflation Rates', 'Retail Sales (in billions)', 'GDP (in trillions)']:
    descriptive_stats.loc[(column, 'mode'), :] = (
        data.groupby('Year')[column].apply(lambda x: x.mode()[0])
    )

descriptive_stats
```

Out[3]:

	Year	2019	2020
Unemployment Rates	mean	3.675000	8.091667
	std	0.135680	3.625781
	var	0.018409	13.146288
	min	3.500000	3.500000
	median	3.600000	7.300000
	max	4.000000	14.800000
Inflation Rates	mean	1.816667	1.233333
	std	0.228963	0.706249
	var	0.052424	0.498788
	min	1.500000	0.100000
	median	1.800000	1.250000
	max	2.300000	2.500000
Retail Sales (in billions)	mean	450.234167	463.329167
	std	5.921782	33.957943
	var	35.067499	1153.141863
	min	439.790000	375.920000
	median	450.520000	476.710000
	max	458.920000	492.700000
GDP (in trillions)	mean	21.520000	21.323333
	std	0.324654	1.011366
	var	0.105400	1.022861
	min	21.080000	18.940000
	median	21.570000	21.765000
	max	21.990000	22.130000
Unemployment Rates	iqr	0.125000	4.275000
Inflation Rates	iqr	0.250000	0.525000
Retail Sales (in billions)	iqr	7.075000	34.750000
GDP (in trillions)	iqr	0.510000	1.047500
Unemployment Rates	mode	3.600000	6.700000
Inflation Rates	mode	1.800000	1.200000
Retail Sales (in billions)	mode	439.790000	375.920000
GDP (in trillions)	mode	21.080000	20.980000

▼ Mean Confidence Intervals for 2019

```
In [6]: import pandas as pd
import scipy.stats as stats

data = pd.read_csv('EconomyData.csv')

data['Year'] = pd.to_datetime(data['Months (2019 - 2020)']).dt.year

#All confidence intervals for the mean were calculated by changing the indicators.
data_2019 = data[data['Year'] == 2019]['GDP (in trillions)']

mean = data_2019.mean()
sem = stats.sem(data_2019)

t_critical = stats.t.ppf(q = 0.975, df = len(data_2019) - 1)

margin_error = t_critical * sem

ci_lower = round((mean - margin_error), 6)
ci_upper = round((mean + margin_error), 6)

(ci_lower, ci_upper)
```

Out[6]: (21.313725, 21.726275)

▼ Mean Confidence Intervals for 2020

```
In [7]: import pandas as pd
import scipy.stats as stats

data = pd.read_csv('EconomyData.csv')

data['Year'] = pd.to_datetime(data['Months (2019 - 2020)']).dt.year

#All confidence intervals for the mean were calculated by changing the indicators.
data_2020 = data[data['Year'] == 2020]['GDP (in trillions)']

mean = data_2020.mean()
sem = stats.sem(data_2020)

t_critical = stats.t.ppf(q = 0.975, df = len(data_2020) - 1)

margin_error = t_critical * sem

ci_lower = round((mean - margin_error), 6)
ci_upper = round((mean + margin_error), 6)

(ci_lower, ci_upper)
```

Out[7]: (20.680742, 21.965924)

▼ Variance Confidence Intervals for 2019

```
In [8]: import pandas as pd
import scipy.stats as stats

data = pd.read_csv('EconomyData.csv')

data['Year'] = pd.to_datetime(data['Months (2019 - 2020)']).dt.year

#All confidence intervals for the variance were calculated by changing the indicators.
data_2019 = data[data['Year'] == 2019]['GDP (in trillions)']

sample_variance = data_2019.var()

chi2_lower = stats.chi2.ppf(q=0.025, df=len(data_2019) - 1)
chi2_upper = stats.chi2.ppf(q=0.975, df=len(data_2019) - 1)

ci_lower_var = round((((len(data_2019) - 1) * sample_variance) / chi2_upper, 6)
ci_upper_var = round((((len(data_2019) - 1) * sample_variance) / chi2_lower, 6)

print(ci_lower_var, ci_upper_var)
```

0.052892 0.303846

▼ Variance Confidence Intervals for 2020

```
In [9]: import pandas as pd
import scipy.stats as stats

data = pd.read_csv('EconomyData.csv')

data['Year'] = pd.to_datetime(data['Months (2019 - 2020)']).dt.year

#All confidence intervals for the variance were calculated by changing the indicators.
data_2020 = data[data['Year'] == 2020]['GDP (in trillions)']

sample_variance = data_2020.var()

chi2_lower = stats.chi2.ppf(q=0.025, df=len(data_2020) - 1)
chi2_upper = stats.chi2.ppf(q=0.975, df=len(data_2020) - 1)

ci_lower_var = round((((len(data_2020) - 1) * sample_variance) / chi2_upper, 6)
ci_upper_var = round((((len(data_2020) - 1) * sample_variance) / chi2_lower, 6)

print(ci_lower_var, ci_upper_var)
```

0.513296 2.948692

▼ Mean Significance Test

```
In [10]: import pandas as pd
         from scipy import stats

         data = pd.read_csv('EconomyData.csv')

         data['Year'] = pd.to_datetime(data['Months (2019 - 2020)']).dt.year

         #All Test for Significance for mean were calculated by changing the indicators.
         data_2019 = data[data['Year'] == 2019]['GDP (in trillions)']
         data_2020 = data[data['Year'] == 2020]['GDP (in trillions)']

         t_stat, _ = stats.ttest_ind(data_2019, data_2020, equal_var=True)

         df = min(len(data_2019) - 1, len(data_2020) - 1)

         alpha = 0.05
         t_critical = stats.t.ppf(1 - alpha / 2, df)

         print(f'T-statistic: {t_stat}')
         print(f'Critical t-value: {t_critical}')

         if abs(t_stat) > t_critical:
             print("Reject the null hypothesis")
         else:
             print("Fail to reject the null hypothesis")

T-statistic: 0.6413818523568051
Critical t-value: 2.200985160082949
Fail to reject the null hypothesis
```

▼ Variance Significance Test

```
In [11]: import pandas as pd
         from scipy import stats

         data = pd.read_csv('EconomyData.csv')

         data['Year'] = pd.to_datetime(data['Months (2019 - 2020)']).dt.year

         #All Test for Significance for variance were calculated by changing the indicators.
         data_2019 = data[data['Year'] == 2019]['GDP (in trillions)']
         data_2020 = data[data['Year'] == 2020]['GDP (in trillions)']

         f_stat = data_2020.var(ddof=1) / data_2019.var(ddof=1)

         df1 = len(data_2020) - 1
         df2 = len(data_2019) - 1

         alpha = 0.05
         f_critical = stats.f.ppf(1 - alpha / 2, df1, df2)

         print(f'F-statistic: {f_stat}')
         print(f'Critical F-value: {f_critical}')

         if f_stat > f_critical:
             print(f"Reject the null hypothesis")
         else:
             print(f"Fail to reject the null hypothesis")

F-statistic: 9.704559829797013
Critical F-value: 3.473699051005809
Reject the null hypothesis
```

Simple Linear Regression

```
In [6]: import pandas as pd
import statsmodels.api as sm
import matplotlib.pyplot as plt

data = pd.read_csv('EconomyData.csv')

X = data['Retail Sales (in billions)']
y = data['GDP (in trillions)']

X = sm.add_constant(X)

model = sm.OLS(y, X).fit()

print(model.summary())

plt.scatter(data['Retail Sales (in billions)'], data['GDP (in trillions)'], label='Data points')
plt.plot(data['Retail Sales (in billions)'], model.predict(X), color='red', label='Regression line')
plt.xlabel('Retail Sales (in billions)')
plt.ylabel('GDP (in trillions)')
plt.legend()
plt.show()
```

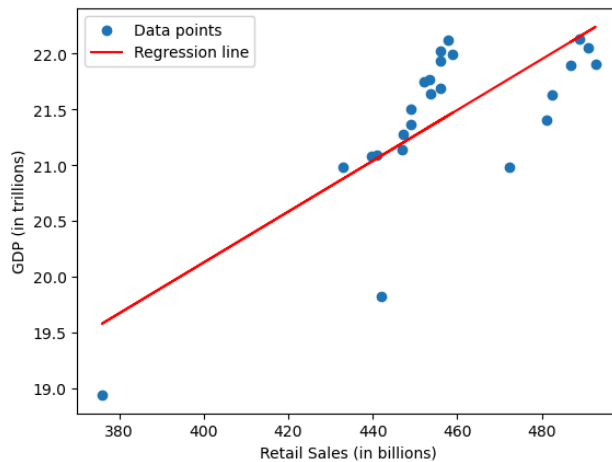
```
=====
                        OLS Regression Results
=====
Dep. Variable:      GDP (in trillions)    R-squared:                0.579
Model:              OLS                  Adj. R-squared:           0.560
Method:             Least Squares        F-statistic:             30.31
Date:               Fri, 12 Apr 2024      Prob (F-statistic):       1.57e-05
Time:               01:59:28             Log-Likelihood:          -15.970
No. Observations:   24                  AIC:                    35.94
Df Residuals:       22                  BIC:                    38.30
Df Model:            1
Covariance Type:    nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	11.0099	1.894	5.813	0.000	7.082	14.938
Retail Sales (in billions)	0.0228	0.004	5.505	0.000	0.014	0.031

```
=====
Omnibus:            4.107    Durbin-Watson:           0.328
Prob(Omnibus):      0.128    Jarque-Bera (JB):         2.693
Skew:               -0.808    Prob(JB):                 0.260
Kurtosis:           3.281    Cond. No.                  8.63e+03
=====
```

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 8.63e+03. This might indicate that there are strong multicollinearity or other numerical problems.



Multiple Linear Regression

```
In [2]: import pandas as pd
import statsmodels.api as sm

data = pd.read_csv('EconomyData.csv')

X = data[['Retail Sales (in billions)', 'Unemployment Rates', 'Inflation Rates']]
X = sm.add_constant(X)

y = data['GDP (in trillions)']

model = sm.OLS(y, X).fit()

print(model.summary())
```

```
=====
                        OLS Regression Results
=====
Dep. Variable:      GDP (in trillions)    R-squared:                0.955
Model:              OLS                  Adj. R-squared:            0.948
Method:             Least Squares         F-statistic:              140.5
Date:               Fri, 12 Apr 2024      Prob (F-statistic):       1.32e-13
Time:               00:53:05              Log-Likelihood:           10.773
No. Observations:   24                   AIC:                     -13.55
Df Residuals:       20                   BIC:                     -8.833
Df Model:            3
Covariance Type:    nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	11.5400	0.732	15.758	0.000	10.012	13.068
Retail Sales (in billions)	0.0204	0.001	14.153	0.000	0.017	0.023
Unemployment Rates	-0.0423	0.025	-1.723	0.100	-0.094	0.009
Inflation Rates	0.5473	0.140	3.911	0.001	0.255	0.839

```
=====
Omnibus:                2.098    Durbin-Watson:                1.057
Prob(Omnibus):           0.350    Jarque-Bera (JB):                1.548
Skew:                    -0.614    Prob(JB):                        0.461
Kurtosis:                2.801    Cond. No.                      9.73e+03
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 9.73e+03. This might indicate that there are strong multicollinearity or other numerical problems.